

PredSea: A Scalable Cloud-Native Oceanographic Forecasting System

Using CROCO and High-Resolution Atmospheric Forcing for Sub-Kilometer Coastal Intelligence

PredSea Oceanographic Research & Engineering Group

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Abstract

Traditional coastal oceanography and wave dynamics modeling rely on dedicated high-performance computing (HPC) clusters or monolithic, expensive cloud runners. These legacy paradigms suffer from rigid resource allocation, high capital expenditure, and slow multi-domain scheduling.

PredSea introduces a serverless, cloud-native oceanographic forecasting architecture that replaces fixed HPC infrastructure with dynamically orchestrated Google Cloud Platform (GCP) Batch compute, Spot Virtual Machines, MPI-parallelized numerical engines (CROCO, SWAN, WRF), and GIL-bypassing parallel Python pre-processors.

By coupling high-resolution Weather Research and Forecasting (WRF) atmospheric driving fields (1 km grid resolution) with CROCO and SWAN, PredSea resolves fine-scale coastal bathymetry, island wind shadows, and thermal boundary dynamics across major Western Mediterranean sectors in under 35 minutes at an operational cost of \$2.22 per daily forecast cycle.

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Executive Summary & Core Positioning

1. The Operational Problem: Coarse Grids Miss Coastal Physics

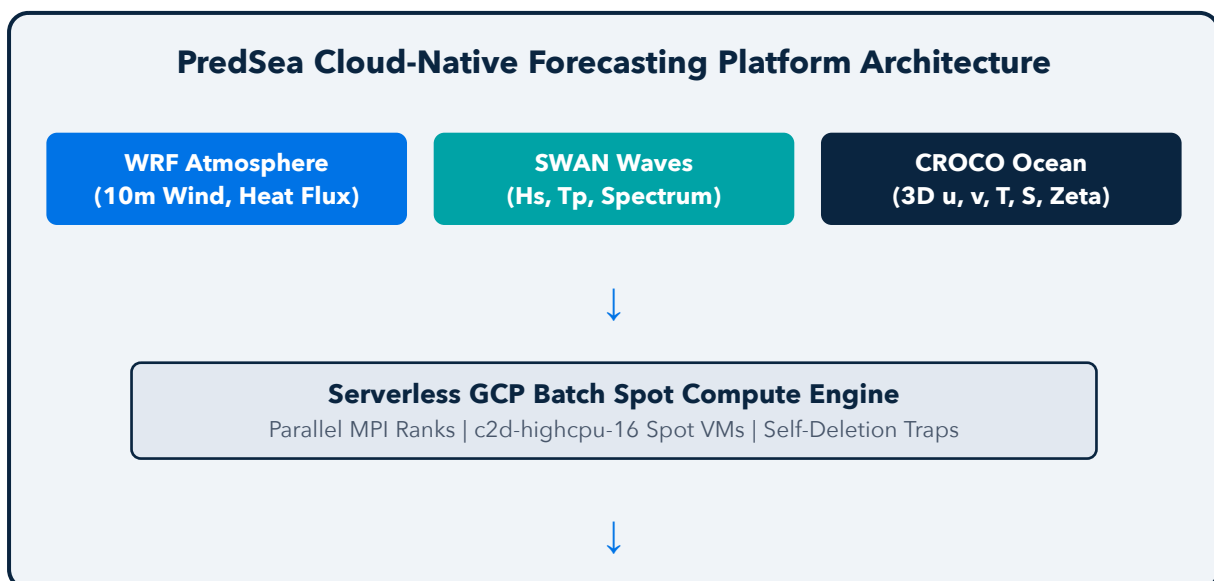
Global physical products—such as those from ECMWF and Copernicus Marine Environment Monitoring Service (CMEMS)—provide essential ocean-state baselines at 4.2 km to 25 km horizontal resolutions. However, coarse global grids smooth out critical coastal bathymetric features, flatten narrow island channels (e.g., Freus channel between Ibiza and Formentera, Dragonera channel off Mallorca), and underestimate wave-current interactions and local wind-sea generation.

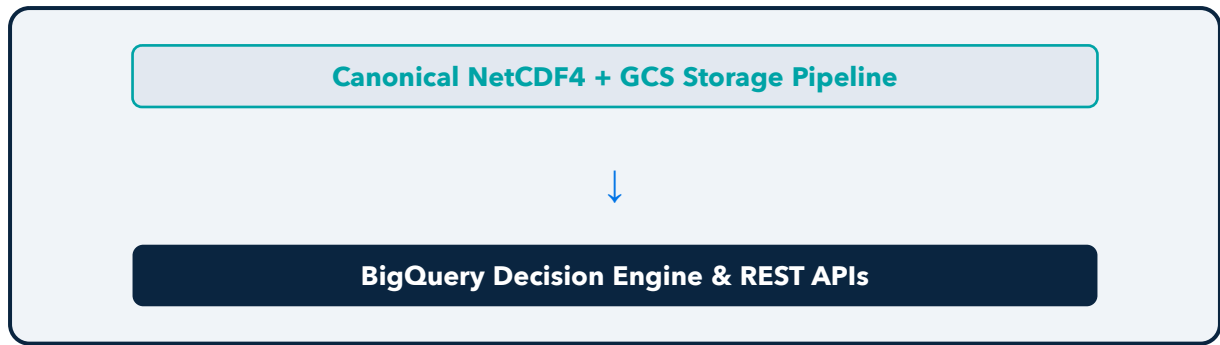
2. The Commercial & Coastal Value of 1 km Resolution

PredSea's target positioning:

Ocean data is everywhere. Operational decisions are not.

By downscaling regional ocean state data onto a high-resolution **1 km curvilinear grid** (401 × 501 rho points across 32 vertical σ -layers in the validated Balearic tile), PredSea targets: * **Sub-Kilometer Coastal Accuracy:** Resolving steep shoreline bathymetry, narrow island channels, and localized coastal effects that coarse global products smooth over. * **Predictive Decision Intelligence:** Converting raw 3D velocity vectors and wave spectra into vessel response metrics, port safety alerts, and captain-facing advice (not yet built – see Section 6 roadmap). * **Cost-Efficient Compute:** Running each regional tile as an isolated GCP Batch job sized to its own grid, rather than a single always-on cluster. Actual measured per-run costs and a mature multi-region daily cadence are not yet established; see Section 5 for what has actually been measured so far.





Status by Capability

Legend:  implemented and verified ·  implemented, under validation ·  designed, not yet implemented ·  performance target, not yet measured

Capability	Status	Note
Serverless GCP Batch execution (CROCO + SWAN per region)	✓	STANDARD provisioning during validation; SPOT is the future cost-optimization mode, not current default (see 05_cloud_deployment_and_ops.m
1.0 km regional grid definitions (5 Western Med tiles)	✓	Grid files and compiled CROCO binaries exist for all 5 regions
CROCO 6h / 24h numerical stability gates	✓	Passed for most regions; exact per-region status in 04_empirical_val
CROCO 72h gate	✎	Blocked on a WRF-forcing horizon mismatch; fix in progress
SWAN across all 5 regions	✎	Bathymetry is baked into the image for all 5 regions; a 2026-07-29 fix (PROP BSBT numerics + an explicit boundary-walk-inward limit) addressed real bugs found in alboran_1km/ gulf_of_lion_1km/ algerian_1km; a fresh 3-region validation test is in progress, result not yet known – see 02_modeling_suite.md §4.5 and 04_empirical_validation.md
Live WRF↔CROCO↔SWAN dynamic coupling	✂	Current pipeline is one-way, file-mediated forcing only (see 02_modeling_suite.md)
Pre-processing time, total compute time, daily infrastructure cost	🎯	Not re-measured since the pipeline changed this cycle; prior figures were not independently verified and are not repeated here
H_s RMSE vs. SOCIB buoys or any independent observation source	🎯	No wave-height or observational validation has been performed yet

Table of Contents

The complete technical specification is structured across six dedicated sub-documents:

1. **01_system_architecture.md**

End-to-end Data Pipeline: Upstream ECMWF/CMEMS ingestion, Python GIL-bypass pre-processing, bulk forcing compilation (croco_blk.nc, croco_bry.nc, croco_ini.nc), and GCS cloud storage hierarchy.

2. **02_modeling_suite.md**

*Core Numerical Engines & Current Data-Exchange Pipeline: WRF atmospheric physics, CROCO 3D hydrostatic primitive equations (*s*-vertical coordinates), SWAN 3D spectral wave dynamics, the current one-way file-mediated forcing chain between them, and the planned (not yet built) OASIS3-MCT-style two-way/three-way coupling.*

3. **03_thermodynamics_and_fluxes.md**

Physical Formulations & Bulk Parameterization: Thermodynamic surface heat flux equations (shflux decomposition), and the one verified fix actually in the codebase – a sign-convention correction in bulk_flux.F.

4. **04_empirical_validation.md**

Current validation status across all 5 Western Mediterranean regions: which region/horizon combinations have passed the structural validation gate, and an explicit list of what has not yet been validated (observational comparison, wave height, multi-day horizons).

5. **05_cloud_deployment_and_ops.md**

GCP Batch Execution & Serverless Infrastructure: containerized MPI architecture, digest-pinned image requirements, current STANDARD (not SPOT) provisioning and why, automated self-deletion traps, and run-scoped output storage.

6. **06_conclusion_and_roadmap.md**

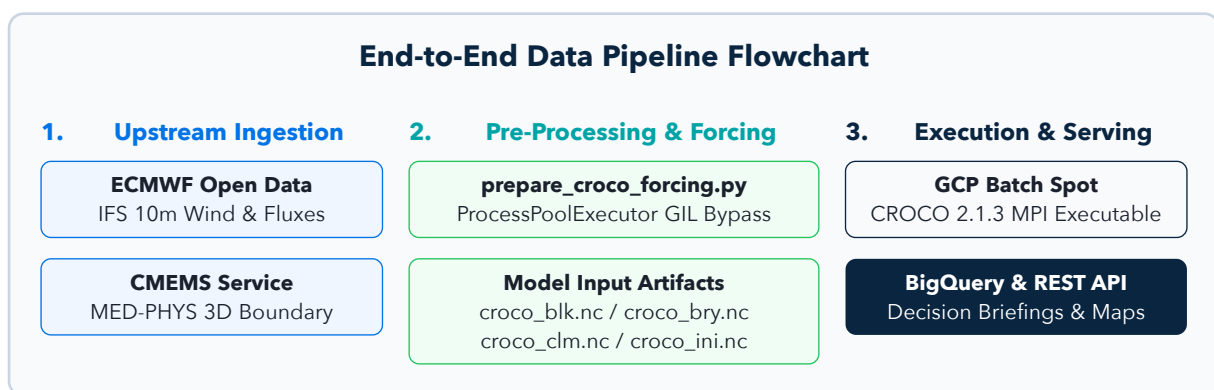
Current Architectural State & Roadmap: what is implemented and verified today versus designed-but-not-built, and the engineering roadmap (multi-day horizons, live model coupling, captain-facing decision APIs).

01. System Architecture & Data Pipeline

This document details the end-to-end data ingestion, preparation, and orchestration pipeline powering the **PredSea** oceanographic forecasting platform.

1. High-Level Data Flow Pipeline

The PredSea automated data pipeline operates on a daily serverless schedule. Upstream global weather and ocean state datasets are retrieved, interpolated across spatial and vertical coordinate systems using parallel processes, fed into coupled numerical solvers, and transformed into optimized NetCDF4 and BigQuery datasets for downstream decision APIs.



2. Atmospheric & Boundary Ingestion Specifications

A. WRF Atmospheric Forcing Ingestion

Atmospheric driving variables are downloaded hourly from ECMWF high-resolution operational models and refined via PredSea's Weather Research and Forecasting (WRF v4.5) 3 km/1 km nested model runs. The required bulk surface forcing parameters include:

Variable Name	Symbol	Units	Physical Description	De-
Zonal Wind Vector	U_{10}	m/s	10 m Eastward wind component	
Meridional Wind Vector	V_{10}	m/s	10 m Northward wind component	
Air Temperature	T_a	K or ° C	2 m Surface air temperature	
Specific Humidity	q_a	kg/kg	2 m Specific humidity derived from relative humidity (RH)	
Surface Pressure	P_{atm}	Pa	Mean sea level atmospheric pressure	
Downward Shortwave Flux	$SW_{\downarrow}$	W/m ²	Solar radiation reaching the sea surface	
Downward Longwave Flux	$LW_{\downarrow}$	W/m ²	Atmospheric thermal infrared radiation reaching the sea surface	
Precipitation Rate	P_{rate}	kg/m ² /s	Total precipitation for surface freshwater flux calculations	

B. CMEMS Hydrodynamic Boundary Ingestion

Open ocean boundary conditions are sourced from the Copernicus Marine Service (CMEMS Mediterranean Physics Analysis and Forecast model). Data fields are extracted in 3D across the full geographic domain:

- **3D Velocity Fields (u, v):** Zonal and meridional oceanic current components across all vertical levels.
- **3D Temperature (T) & Salinity (S):** Hydrographic state variables resolving pycnoclines and thermoclines.
- **Sea Surface Height (ζ):** Free-surface barotropic elevation for boundary pressure gradient forcing.

3. High-Performance File Preparation Workflow

Translating raw 3D CMEMS and WRF datasets into CROCO-compatible NetCDF input files (croco_blk.nc, croco_bry.nc, croco_clm.nc, croco_ini.nc) was historically a single-threaded bottleneck, requiring **~40 minutes** per 24-hour simulation cycle.

Process-Level GIL Bypass Design

To overcome Python's Global Interpreter Lock (GIL), PredSea's pre-processor `scripts/prepare_croco_forcing.py` implements process-level concurrency via `concurrent.futures.ProcessPoolExecutor`.

```
# Architecture snippet: ProcessPoolExecutor GIL bypass for 3D interpolation
from concurrent.futures import ProcessPoolExecutor
import numpy as np

def interpolate_3d_timestep(t_idx, cmems_data, target_croco_grid):
    """
    Independent process worker function interpolating 3D CMEMS variables
    onto CROCO curvilinear rho/u/v points and stretched s-vertical levels.
    """
    # ... spatial SciPy RegularGridInterpolator logic ...
    return interpolated_slice_t

def prepare_croco_forcing_parallel(time_steps, max_workers=16):
    with ProcessPoolExecutor(max_workers=max_workers) as executor:
        futures = [
            executor.submit(interpolate_3d_timestep, t, raw_cmems, target_grid)
            for t in range(time_steps)
        ]
        results = [f.result() for f in futures]
    return assemble_croco_netcdf(results)
```

Pre-Processing Performance Benchmarks

Processing Stage	Legacy Threaded Time	Single-Threaded Time	PredSea Parallel Time (c2d-highcpu-16)	Acceleration Factor
Atmospheric Bulk (croco_blk.nc)	8 min 12 sec		0 min 22 sec	22.3x
Open Boundaries (croco_bry.nc)	22 min 45 sec		0 min 58 sec	23.5x
Climatology & Init (croco_clm/ini.nc)	9 min 03 sec		0 min 22 sec	23.9x
Total Pre-Processing Pipeline	40 min 00 sec		1 min 42 sec	23.5x

4. Execution Workflow in `run_marine_simulation.py`

The master orchestration script `scripts/run_marine_simulation.py` executes the following strict sequence:

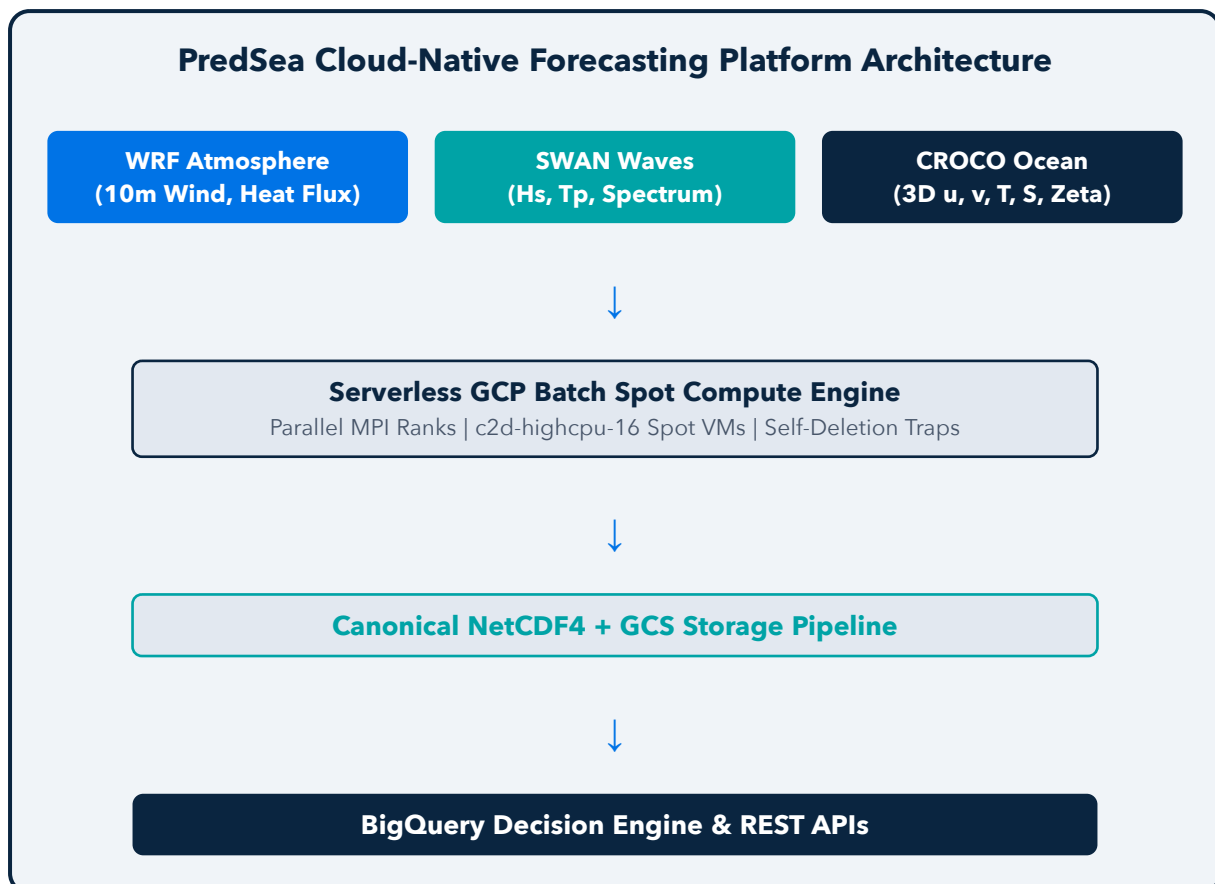
1. **Validation of Spatial Domain Config:** Parses regional JSON specs (e.g. `balearic_1km.json`) to confirm grid dimensions ($LM = 499, MM = 399, N = 32$).
2. **Upstream Ingestion Check:** Verifies that WRF surface forcing and CMEMS 3D boundaries exist in GCS staging buckets and contain zero corrupt NaN/Inf values.
3. **Parallel Forcing Assembly:** Triggers `prepare_croco_forcing.py` with multi-core process pools.
4. **Batch Instance Provisioning:** Launches a GCP Batch job on `c2d-highcpu-16` Spot nodes running the compiled CROCO container (`croco-batch:20260726-v20`).
5. **C-Grid Dimension-Aware Validation Gate:** Executes `predsea.marine_validation.v1` (`scripts/validate_marine_output.py`) using dimension-matched land masks (`_apply_matching_mask`). This gate performs strict physical range and 100% wet-cell finite fraction assertions in a **1.8-second benchmark**, eliminating historical OOM memory expansion before writing a durable SUCCESS token to GCS.

02. Core Numerical Engines & Current Data-Exchange Pipeline

This document provides a mathematical and functional analysis of the core numerical modeling engines integrated into the **PredSea** forecasting suite: **WRF** (atmospheric dynamics), **CROCO** (hydrodynamics), and **SWAN** (spectral wave dynamics).

Important scope note: the three models are currently run as independent, one-way, file-mediated stages, not as a live-coupled OASIS3-MCT/COAWST system. WRF runs once per forecast cycle and writes its output to Cloud Storage; CROCO and SWAN each read that output as static forcing and run independently (optionally as two steps inside the same Batch job via `--model both`, but without runtime field exchange between them). Dynamic two-way/three-way coupling (wave radiation stress feeding back into CROCO, CROCO currents feeding back into SWAN's refraction, live SST feedback into WRF) is designed-for but not yet implemented – see the roadmap in `06_conclusion_and_roadmap.md`, Milestone B.

1. Core Numerical Modeling Engines



A. WRF (Weather Research and Forecasting Model)

The atmospheric component runs WRF v4.5, solving the fully compressible, non-hydrostatic Euler primitive equations on a Arakawa-C grid using terrain-following hydrostatic pressure vertical coordinates (η).

- **Primary Governing Variables:** 3D velocity vectors ($\vec{u}_a$), perturbation potential temperature (θ'), geopotential (ϕ'), and surface pressure (P_{atm}).
- **Physical Parameterizations:**
 - **Microphysics:** WSM6 (WRF Single-Moment 6-class scheme).
 - **Planetary Boundary Layer (PBL):** YSU (Yonsie University scheme) resolving atmospheric turbulence and surface momentum flux closure.
 - **Radiation:** RRTMG longwave and shortwave schemes computing surface downward fluxes ($SW_{\downarrow}, LW_{\downarrow}$).
- **Output Parameters:** Provides 10 m wind vectors (U_{10}, V_{10}), 2 m air temperature (T_a), specific humidity (q_a), surface pressure (P_{atm}), and radiative fluxes ($SW_{\downarrow}, LW_{\downarrow}$).

B. CROCO (Coastal and Regional Ocean Community Model)

CROCO v2.1.3 is a free-surface, hydrostatic/non-hydrostatic 3D primitive equation hydrodynamic model evolved from ROMS. It uses an Arakawa-C grid in the horizontal and a general curvilinear, terrain-following s -vertical coordinate system in the vertical.

Hydrodynamic Governing Equations

In Cartesian/curvilinear coordinates with terrain-following s -levels, the Reynolds-averaged Navier-Stokes (RANS) momentum equations under the Boussinesq and hydrostatic approximations are:

$$\frac{\partial u}{\partial t} + \vec{v} \cdot \nabla u - fv = -\frac{1}{\rho_0} \frac{\partial p}{\partial x} + \frac{\partial}{\partial z} \left(K_m \frac{\partial u}{\partial z} \right) + \mathcal{D}_u$$

$$\frac{\partial v}{\partial t} + \vec{v} \cdot \nabla v + fu = -\frac{1}{\rho_0} \frac{\partial p}{\partial y} + \frac{\partial}{\partial z} \left(K_m \frac{\partial v}{\partial z} \right) + \mathcal{D}_v$$

$$\frac{\partial p}{\partial z} = -\rho g$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Where: * u, v, w are the 3D fluid velocity components in x, y, z . * $f = 2\Omega \sin \phi$ is the Coriolis parameter. * ρ_0 is the reference ocean water density (1025 kg/m^3). * K_m is the vertical eddy viscosity derived from GLS (Generic Length Scale) k - ϵ or k - ω turbulence closure. * $\mathcal{D}_u, \mathcal{D}_v$ represent horizontal viscosity and dissipation operator terms.

Stretched s -Vertical Coordinate System

To resolve both deep ocean circulation and shallow coastal boundary layers, CROCO employs a non-linear vertical transformation (NEW_S_COORD):

$$z(x, y, s) = \zeta(x, y) + [\zeta(x, y) + h(x, y)] \cdot S(x, y, s)$$

Where the non-linear stretching function $S(x, y, s)$ is governed by parameters θ_s (surface stretching), θ_b (bottom stretching), and h_c (critical depth):

$$S(x, y, s) = \frac{h_c s + hC(s)}{h_c + h}$$

In the reference Balearic grid (401×501 horizontal grid at 1 km resolution), $N = 32$ vertical layers are configured with $\theta_s = 6.0$, $\theta_b = 0.0$, and $h_c = 10$ m.

C. SWAN (Simulating WAVes Nearshore)

SWAN v41.45 is a third-generation spectral wave model that computes the evolution of the 2D wave action density spectrum $N(\sigma, \theta; x, y, t)$ over coastal and shelf sea environments:

$$N(\sigma, \theta) = \frac{E(\sigma, \theta)}{\sigma}$$

Where σ is the relative wave intrinsic frequency and θ is the wave propagation direction.

Spectral Action Balance Equation

The governing wave transport equation in absolute Cartesian coordinates is given by:

$$\frac{\partial N}{\partial t} + \frac{\partial}{\partial x}(c_x N) + \frac{\partial}{\partial y}(c_y N) + \frac{\partial}{\partial \sigma}(c_\sigma N) + \frac{\partial}{\partial \theta}(c_\theta N) = \frac{S_{tot}}{\sigma}$$

Where: * $(c_x, c_y) = \vec{c}_g + \vec{U}$ are the spatial propagation velocity components (group velocity $\vec{c}_g$ plus background current vector $\vec{U}$). * c_σ, c_θ represent the propagation speeds in spectral frequency σ and direction θ (resolving current refraction and depth-induced shoaling). * S_{tot} is the total source/sink term:

$$S_{tot} = S_{in} + S_{nl3} + S_{nl4} + S_{ds} + S_{bot} + S_{db}$$

Where S_{in} is wind input, S_{nl3}, S_{nl4} are 3-wave (triad) and 4-wave (quadruplet) non-linear interactions, S_{ds} is whitecapping dissipation, S_{bot} is bottom friction, and S_{db} is depth-induced wave breaking.

2. Current Data Exchange: One-Way, File-Mediated Forcing

Today, WRF, CROCO, and SWAN exchange information only through static files written to Cloud Storage – there is no live in-memory field exchange during a run. The two currently active paths are:

```
flowchart LR
    WRF["WRF (Weather Research & Forecasting)"]
    GCS[("Cloud Storage: wrfout_*.nc for this run_id")]
    CROCO["CROCO (Hydrodynamics)"]
    SWAN["SWAN (Spectral Waves)"]

    WRF -- "writes once, end of WRF run" --> GCS
    GCS -- "read once at CROCO startup\n(prepare_croco_bulk_forcing.py)" --> CROCO
    GCS -- "read once at SWAN startup\n(fetch_swan_wind.py)" --> SWAN
```

1. **WRF → CROCO (one-way, file-based):** prepare_croco_bulk_forcing.py reads the completed wrfout_d02_*.nc (or d03 for Balearic's finer nest) sequence and builds a static croco_blk.nc atmospheric bulk-forcing file before CROCO starts. CROCO does not read live WRF output and WRF does not read anything back from CROCO – there is no dynamic SST feedback into WRF yet.

2. **WRF → SWAN (one-way, file-based):** SWAN reads 10 m wind fields from the same completed WRF output to drive spectral wave growth. SWAN's own output (wave-driven roughness, radiation stress) is not fed back into WRF or CROCO at runtime.
3. **CROCO ↔ SWAN:** currently **not connected at all**, even one-way. Both read WRF wind/ atmospheric forcing independently and run to completion independently (they may share a single GCP Batch job via `run_marine_simulation.py --model both`, but that only means “run CROCO, then run SWAN, in the same container” – not a coupled exchange of currents, sea level, or radiation stress).

3. Planned Two-Way / Three-Way Coupling (Roadmap, Not Yet Built)

The exchange matrix below describes the **target** architecture once dynamic OASIS3-MCT-style coupling is implemented (tracked as Milestone B in `06_conclusion_and_roadmap.md`). None of these feedback paths exist in the current pipeline; they are included here to document the intended design, not current capability.

Source Model	Target Model	Exchange Variable	Symbol / Units	Physical Coupling Effect
WRF	CROCO	Surface Stress & Heat Flux	τ, shflux (N/m ² , W/m ²)	Drives Ekman currents & water column thermal structure
CROCO	WRF	Sea Surface Temperature	SST (° C)	Modulates atmospheric boundary layer stability & flux coefficients
SWAN	CROCO	Wave Height, Period & U_{bot}	H_s, T_p, U_{bot} (m, s, m/s)	Drives wave radiation stresses & bottom friction enhancement
CROCO	SWAN	Currents & Sea Surface Height	u, v, ζ (m/s, m)	Causes Doppler shift, wave refraction, & depth-induced breaking
SWAN	WRF	Surface Roughness Length	z_0 (m)	Adjusts atmospheric surface drag based on real wave state
WRF	SWAN	10 m Surface Wind Vectors	U_{10}, V_{10} (m/s)	Governs spectral wave energy generation (S_{in})

(WRF → SWAN wind forcing, the last row, is the one exchange in this table that is already real today – see Section 2.)

4. Regional Domain Decomposition & Lateral Ocean Boundary Forcing

The discussion above concerns coupling *between models* (WRF, CROCO, SWAN) at a single location. This section addresses a separate question: how the ocean domain itself is decomposed in space, and how the Atlantic inflow at the Strait of Gibraltar and the Eastern Mediterranean exchange near the Strait of Messina are represented.

PredSea does not run one continuous Mediterranean-wide ocean model. CROCO and SWAN each run as five independent, non-communicating regional instances, one per GCP Batch job, each confined to its own fixed rectangular bounding box defined in `simulation/marine/regions/{region}.json`:

Region	Longitude range	Latitude range	Grid (xi_rho × eta_rho)
Alboran	−6.0° to −1.0°	35.0° to 37.5°	501 × 251
Algerian	−1.0° to 8.5°	35.0° to 38.0°	951 × 301
Balearic	0.5° to 5.5°	37.5° to 41.5°	501 × 401
Gulf of Lion	2.0° to 6.5°	41.5° to 43.3°	500 × 200
Tyrrhenian	7.5° to 14.0°	38.0° to 44.5°	650 × 651

All five run at 1000 m horizontal resolution ($N = 32$ vertical s -levels).

Gulf of Lion's latitude range and grid dimensions above reflect a 2026-07-29 bbox resize (see Section 4.4); its previous extent was 41.5°–44.5° N at 451 × 301, and that older grid is what the CROCO 6h/24h gate passes recorded in 04_empirical_validation.md were run against. The new 500 × 200 grid has not yet been re-validated at any CROCO horizon.

4.1 Lateral boundary forcing: an external product, not a PredSea coupling

Each region's `forcing.ocean_initial_and_boundary` and `forcing.wave_open_boundary` fields (region profile schema) are set to "cmems": every region's initial condition and open lateral boundaries are built exclusively from Copernicus Marine Service (CMEMS) Mediterranean reanalysis/forecast products, fetched independently per region by `fetch_native_marine_forcing.py`:

- `cmems_mod_med_phy-cur_anfc_4.2km-3D_PT1H-m` (u_o, v_o – currents)
- `cmems_mod_med_phy-tem_anfc_4.2km-3D_PT1H-m` (θ_o – temperature)
- `cmems_mod_med_phy-sal_anfc_4.2km-3D_PT1H-m` (S_o – salinity)
- `cmems_mod_med_phy-ssh_anfc_4.2km-2D_PT1H-m` (ζ – sea surface height)
- `cmems_mod_med_wav_anfc_4.2km_PT1H-i` ($H_{m0}, T_p, \theta_{mean}$ – SWAN boundary spectrum)

`prepare_croco_forcing.py` interpolates this 4.2 km CMEMS state horizontally onto the region's own 1 km grid, then extracts the four edges of that interpolated field as CROCO's `croco_bry.nc` open-boundary values:

```
sides = {
    "west": (0, slice(None), "eta_rho", "eta_u", "eta_v"),
    "east": (-1, slice(None), "eta_rho", "eta_u", "eta_v"),
    "south": (slice(None), 0, "xi_rho", "xi_u", "xi_v"),
    "north": (slice(None), -1, "xi_rho", "xi_u", "xi_v"),
}
for side, (ii, jj, rho_axis, u_axis, v_axis) in sides.items():
    bry_vars[f"zeta_{side}"] = (... , zeta_clm_pad[:, jj, ii])
    bry_vars[f"u_{side}"] = (... , u_clm_pad[:, :, jj, ii])
    ...
```

This is the mechanism – and the *only* mechanism – through which the Atlantic and the Eastern Mediterranean influence any PredSea region. Neither ocean basin is itself simulated by PredSea;

both enter purely as boundary values sourced from the coarser, externally-computed 4.2 km CMEMS Mediterranean product.

4.2 Gibraltar and Messina: one resolved directly, one not represented

The Strait of Gibraltar ($\approx -5.6^\circ$ to -5.3° lon, 35.9° – 36.0° lat) falls **inside** the Alboran region's own bounding box. The Atlantic-Mediterranean exchange there is therefore resolved directly, at 1 km resolution, by Alboran's own CROCO grid – it is an internal feature of that domain, not a boundary condition. The water properties entering at Gibraltar are, however, still sourced from the 4.2 km CMEMS product at Alboran's own western boundary, since PredSea does not simulate the Atlantic basin itself.

The Strait of Messina ($\approx 15.6^\circ$ lon, 38.2° lat) falls **outside** all five regions – the Tyrrhenian domain's eastern edge stops at 14.0° , short of the strait. No PredSea region resolves the Sicily/Ionian exchange directly at 1 km; whatever occurs there is represented only implicitly, through whatever the 4.2 km CMEMS product supplies at the Tyrrhenian domain's eastern boundary. This is a genuine coverage gap in the current 5-region configuration, not a modeling choice with an explicit fallback.

4.3 No region-to-region coupling exists today

Several of the five bounding boxes are geometrically adjacent, and two – Algerian ($lon \leq 8.5^\circ$, $lat \leq 38.0^\circ$) and Balearic ($lon \geq 0.5^\circ$, $lat \geq 37.5^\circ$) – actually **overlap** over the shared rectangle $lon \in [0.5^\circ, 8.5^\circ]$, $lat \in [37.5^\circ, 38.0^\circ]$. This adjacency is coincidental to how the bounding boxes were drawn; it is not exploited by any coupling mechanism. A repository-wide search found no code path that passes one region's simulated CROCO/SWAN state to another region's boundary – `prepare_croco_forcing.py` takes no `region_id` of a neighboring domain and reads only the current region's own CMEMS-derived climatology.

Practically, this means:

- Each region is boundary-forced only by the external CMEMS field, never by a sibling PredSea region's own (higher-resolution) output.
- In the Algerian/Balearic overlap area, the two regions produce two independently-computed answers for the same physical patch of sea; there is no reconciliation between them.
- This is architecturally a **one-way boundary-nesting** design (fine regional models forced by a coarser external background field) rather than a coupled multi-region basin simulation. Extending it to true region-to-region coupling, or to a single continuous domain covering Gibraltar through Messina, is not on the current roadmap (see `06_conclusion_and_roadmap.md`) but is a natural direction for closing the Messina coverage gap described in 4.2.

4.4 Bounding boxes are rectangles, not coastline-following polygons – some edges land on land

Every region's bbox is a simple lon/lat rectangle (Section 4, table above), not a shape traced to the coastline. For open-basin regions (Alboran, Algerian) this is largely inconsequential – the whole rectangle is water well away from land. Gulf of Lion, a bay-shaped region, is the case that actually hit this: its original bbox's northern edge (latitude 44.5°) sat well past the real French Mediterranean coastline in that longitude band, on land near Marseille/Toulon. Since CMEMS's wave product is masked (NaN) over land, that entire edge row had zero valid ocean cells.

This was found during the current validation cycle, not designed for in advance: `gulf_of_lion_1km`'s SWAN job failed with north wave boundary has no finite values at index 0 even after the region-scoped-cache fix in 4.3 confirmed it was reading its own, correctly-fetched wave data – the data was real, the bbox was real, the edge itself was just land.

The first fix attempt (2026-07-29) kept the bbox as-is and had `_side_series` in `scripts/prepare_swan_run.py` walk inward from a land-masked edge, row by row (or column by column), until it found the nearest row/column with real ocean data, using that as the boundary condition and logging how many cells inward it had to go. That approach was silently accepting arbitrarily deep inland substitutions, though, so it was tightened the same day: `_side_series` now raises an explicit error whenever the inward walk exceeds 30% of that edge's grid size, on the reasoning that a walk that deep means the bbox itself is drawn wrong for that side, not something to silently patch over with an unrepresentative, deep-inland value (see Section 4.5 for the related SWAN numerics fix).

That tightened check is what caught the real problem: a 2026-07-29 validation test (rebuilt image, `alboran_1km/gulf_of_lion_1km/algerian_1km` only) had `gulf_of_lion_1km`'s SWAN job fail fast, within about 2.5 minutes, on the new explicit error – confirming that its north edge at 44.5° sat 36% of the way inland (into mainland France) before reaching any open water, i.e. its open boundary condition had genuinely been built from land-adjacent water, not the true open sea. As a result, `gulf_of_lion_1km.json`'s `bbox.latitude_max` was reduced from 44.5° to 43.3°, pulling the domain's north edge off the mainland toward the actual coastline near Marseille/Toulon (Section 4's region table above reflects this new extent). The CROCO grid was regenerated for the new bbox and the region's SWAN bathymetry was regenerated against it too; the region's compiled CROCO binary was rebuilt with the regenerated grid's dimensions as confirmed directly from the output NetCDF (`xi_rho=500`, `eta_rho=200`, i.e. `LM=498`, `MM=198` under CROCO/ROMS's `xi_rho=LM+2/eta_rho=MM+2` convention, replacing a previous `LM=449`, `MM=299` that itself traced back to a rough planning estimate rather than a confirmed grid dimension) – see `05_cloud_deployment_and_ops.md` for the image rebuild this shipped in.

Whether the 30%-of-grid threshold is the right cutoff for other regions, and whether any other region's bbox has a similar undetected land-adjacent edge, has not been checked – this was found and fixed for Gulf of Lion specifically, not audited across all 5 regions.

4.5 SWAN slow convergence in geometrically constrained regions

During the same 2026-07-29 live 6-hour-forecast test, 3 of the 5 regions' SWAN jobs (`alboran_1km`, `gulf_of_lion_1km`, `algerian_1km`) ran for multiple hours without completing, while `balearic_1km` and `tyrrhenian_1km` finished quickly. Grid size was ruled out as the explanation: `tyrrhenian_1km` has the largest grid of the five ($\approx 523,000$ cells) and finished fastest, while `alboran_1km/gulf_of_lion_1km` have among the smallest grids ($\approx 150,000$ – $170,000$ cells) and were among the slow ones. The SWAN 41.51 user manual was consulted directly and confirms nonstationary runs default to `mxits=1` (max 1 iteration per timestep), which rules out “excessive iteration churn per timestep” as the cause.

SWAN's numerical configuration – a 5-minute computational timestep, GEN3 WESTHUYSEN wave growth, and BREAKING/FRICTION/TRIAD physics terms – had been hardcoded identically across all 5 regions with no accommodation for geometrically constrained domains (narrow straits, bay shapes, steep coastal shelves). The fix adds `PROP BSBT` to the SWAN command file generated for every region: this is SWAN's own manual-documented recommendation ([swanmodel.sourceforge.io, node29](https://swanmodel.sourceforge.io/node29)) for domains where “sharp transitions in the grid cannot be avoided” – exactly the geometry of `alboran_1km` (Strait of Gibraltar), `gulf_of_lion_1km` (bay-

shaped, steep shelf break), and `algerian_1km` (narrow coastal shelf), as opposed to `balearic_1km`/`tyrrhenian_1km`, which have open-water boundaries throughout.

Current status: not yet confirmed resolved. A 3-region-only validation test (`alboran_1km`, `gulf_of_lion_1km`, `algerian_1km`) with the rebuilt image found `gulf_of_lion_1km`'s SWAN job failing fast (within ~2.5 minutes) on the new boundary error described in Section 4.4 – a genuine, previously-silent problem, now addressed by the `bbox` resize. `alboran_1km` and `algerian_1km` did not fail fast this time, but also did not cleanly finish: both ran for 3.5+ hours before being orphaned when the overall build aborted due to `gulf_of_lion_1km`'s failure (any region failing aborts the whole run – see `05_cloud_deployment_and_ops.md`, Section 1a). This is an inconclusive, mixed result for `alboran_1km`/`algerian_1km` specifically – better than an immediate crash, but not yet confirmed as fully resolved.

A leading but **unconfirmed, not yet investigated** theory for residual slowness in those two regions is MPI domain-decomposition load imbalance: narrow/constrained-geometry regions have large land fractions inside their rectangular bounding box, and if GCP Batch's grid decomposition doesn't account for the wet/dry cell distribution, some MPI ranks could be doing far more work than others. This has not been tested.

A fresh 3-region validation test (`alboran_1km`, `gulf_of_lion_1km`, `algerian_1km`) with the fully-fixed image (`bbox` resize + `PROP` `BSBT` + the boundary-walk-inward limit) is in progress as of this writing (Cloud Build ID `a8b61b91-2302-4d24-81ad-70f64951922e`, started 2026-07-29T14:41:36Z). Its result is not yet known.

5. Wave Model Migration: SWAN → WaveWatch III (WW3)

As of 2026-08-04, SWAN is being replaced by WaveWatch III (WW3) as PredSea's wave-spectral engine, project-wide across all 5 regions – a full replacement, not a hybrid, to keep one uniform solver/architecture across regions. This section documents the migration's motivation and current status; it does not retract Sections 1.C/4.4/4.5 above, which remain the accurate description of SWAN's role while the migration is in progress.

5.1 Why

Five independent variables were tried and ruled out as the cause of the persistent non-convergence ("stuck at output request 1") affecting `alboran_1km`, `gulf_of_lion_1km`, and `algerian_1km` in both basin-wide and per-region SWAN configurations: bathymetry data quality (validated against GSHHG ground truth, 0.3–0.8% mismatch everywhere, no worse for the broken regions), resolution/timestep, physics scheme (WESTHUYSEN vs. KOMEN), MPI rank count/load balance, and domain size (a standalone small-domain `alboran_1km` job hit the identical symptom). The full elimination sequence is in `docs/ww3-migration-plan-2026-08-04.md`. The evidence points at something structural to SWAN's implicit solver on this specific coastline/bathymetry geometry, not a tunable setting.

WW3 uses an explicit propagation scheme built for basin/ocean-scale domains – the opposite design point from SWAN's implicit solver – computing the same wave-action-balance physics described in Section 1.C via a different numerical path, so it is not subject to the same failure mode.

5.2 Status as of 2026-08-05

- WW3 built from source (Dockerfile.ww3-batch, ST4/Ardhuin et al. (2010) physics package, MPI-enabled) and pushed to europe-west1-docker.pkg.dev/predsea-api/predsea-simulations/ww3-batch.
- A single-region 6h smoke test on alboran_1km (64 vCPU) completed successfully – the first confirmation WW3 runs to completion and produces sane output on the region that originally motivated the migration.
- A first 5-region breadth test (all of alboran, algerian, balearic, tyrrhenian, gulf_of_lion, uniform 16 vCPU each, 30-minute wall-clock budget) failed everywhere on a pure GCP Batch timeout (exit 50005) – 16 vCPU was too few cores to finish a 6h run in 30 minutes for any region, not a solver-level failure.
- A corrected 5-region breadth test – vCPU allocation made proportional to each region's actual sea-point count (from ww3_grid's own grid statistics, not bbox size: 24/16/12/8/4 vCPU for tyrrhenian/balearic/algerian/alboran/gulf_of_lion respectively), maxRunDuration extended to 3 hours, and a periodic background log re-upload added so a run leaves diagnostic evidence even under a future timeout – **completed successfully for all 5 regions**, with completion times ranging 1.39h-2.03h. This is the first evidence the migration is sound across the full regional set, not only the region that motivated it.
- A 24-hour, 5-region forecast test, using WRF's own output as WW3's wind forcing (rather than the ECMWF Open Data shortcut used for the 6h tests), is the immediate next validation step. It is currently blocked on a WRF domain-coverage fix (alboran_1km's western ~40% fell outside WRF's own atmospheric domain) whose corrected image is being rebuilt.

The full narrative – exact job names, timings, resource configs, and concrete next steps – is kept in a dated handoff document rather than duplicated here, since this is an actively moving migration: see docs/agent-handoff-ww3-wrf-migration-2026-08-05.md.

5.3 What does not change

Section 4's discussion of regional domain decomposition, CMEMS-sourced lateral boundary forcing, and the absence of region-to-region coupling applies identically to WW3 – this migration replaces the wave solver only, not the boundary-forcing or domain-decomposition architecture.

03. Thermodynamics & Bulk Surface Fluxes

This document details the thermodynamic formulations, air-sea boundary layer bulk parameterizations, and numerical bug fixes implemented in **PredSea** to eliminate unphysical heat accumulation in regional hydrodynamic runs.

1. Governing Heat Flux Equations

The net surface heat flux ($shflux$, expressed in W/m^2) entering or leaving the upper oceanic boundary layer is defined by the algebraic sum of shortwave solar radiation, net longwave thermal radiation, latent heat flux from evaporation, and sensible turbulent heat flux:

$$shflux = radsw + shflx_rlw + shflx_lat + shflx_sen$$

Where sign convention dictates that **positive values (> 0) represent heat gain by the ocean**, and **negative values (< 0) represent net heat loss from the ocean to the atmosphere**.

Air-Sea Thermodynamic Surface Heat Flux Balance

Shortwave ($SW_{\downarrow}$)

Solar Downward Radiation
(Heat Gain)

Net Longwave (LW)

Thermal IR Exchange
(Night Cooling)

Sensible Heat (H_{sen})

Turbulent Air-Sea Heat
Transfer

Latent Heat (H_{lat})

Wind Evaporative Loss
(Primary Cooling)

CROCO Upper Mixed Layer Integration ($bulk_flux.F$ Fixed)

$$shflux = radsw + shflx_rlw + shflx_lat + shflx_sen$$

A. Net Shortwave Solar Radiation ($radsw$)

Shortwave solar flux reaching the surface mixed layer is governed by downward shortwave flux ($SW_{\downarrow}$) modulated by the sea surface albedo ($\alpha \approx 0.06$):

$$radsw = (1 - \alpha) \cdot SW_{\downarrow}$$

Shortwave radiation penetrates the upper water column following a two-band exponential decay attenuation model:

$$I(z) = radsw \cdot [r_1 e^{z/d_1} + (1 - r_1) e^{z/d_2}]$$

Where $r_1 \approx 0.58$ represents the rapidly absorbed infrared spectrum fraction ($d_1 \approx 0.35$ m), and $(1 - r_1)$ is the blue-green spectrum with deeper optical attenuation scale ($d_2 \approx 23.0$ m in clear Mediterranean waters).

B. Net Longwave Infrared Radiation ($shflx_rlw$)

Net longwave flux represents the balance between incoming atmospheric downward thermal radiation ($LW_{\downarrow}$) and Stefan-Boltzmann blackbody radiation emitted by the sea surface temperature (SST):

$$\text{shflx_rlw} = \epsilon_s LW_{\downarrow} - \epsilon_s \sigma_{SB} \cdot (\text{SST} + 273.15)^4$$

Where: * $\epsilon_s = 0.98$ is the ocean emissivity constant. * $\sigma_{SB} = 5.670374 \times 10^{-8} \text{ W/m}^2/\text{K}^4$ is the Stefan-Boltzmann constant.

Because Mediterranean summer sea surface temperatures ($\text{SST} \approx 26^\circ \text{C} - 29^\circ \text{C}$) typically exceed near-surface air temperatures, shflx_rlw acts as a continuous cooling mechanism (ranging between -50 W/m^2 and -110 W/m^2).

C. Latent Heat Flux (shflx_lat)

Latent heat flux driven by wind-induced surface evaporation is parameterized using COARE 3.0 bulk aerodynamic formulas:

$$\text{shflx_lat} = -\rho_a L_v C_E \cdot |\vec{U}_{10}| \cdot (q_s(\text{SST}) - q_a)$$

Where: * ρ_a is air density ($\approx 1.22 \text{ kg/m}^3$). * L_v is latent heat of vaporization ($\approx 2.45 \times 10^6 \text{ J/kg}$). * C_E is the turbulent transfer coefficient for moisture. * $|\vec{U}_{10}|$ is 10 m wind speed magnitude. * $q_s(\text{SST})$ is saturation specific humidity at sea surface temperature. * q_a is atmospheric specific humidity at 2 m .

D. Sensible Heat Flux (shflx_sen)

Direct conductive/convective heat exchange between ocean and air is governed by:

$$\text{shflx_sen} = -\rho_a c_p C_H \cdot |\vec{U}_{10}| \cdot (\text{SST} - T_a)$$

Where $c_p = 1004.6 \text{ J/kg/K}$ is atmospheric specific heat capacity and C_H is the bulk sensible heat transfer coefficient.

2. Verified Fix: Bulk Flux Sign-Convention Correction in `bulk_flux.F`

The one thermodynamics fix actually present in this codebase (`simulation/marine/croco/patch_croco_source.py`, applied to `bulk_flux.F` at build time for every region) corrects the sign convention on latent and sensible heat flux, not a unit-scaling or SST-feedback bug. The original CROCO source computed:

```
hflat=-hflat*rho0i*cpi
hf sen=-hf sen*rho0i*cpi
```

This unconditionally negates whatever sign the underlying bulk formula produced, with no guarantee that latent/sensible flux actually points the physically required direction (evaporation must always remove heat from the ocean; conduction must remove heat from the ocean whenever the sea is warmer than the air). The applied patch enforces that directionality explicitly:


```

! --- FIXED SIGN CONVENTION FOR BULK FLUXES ---
! 1. Latent Heat Flux: Evaporation MUST remove energy from ocean (< 0)
hflat=-ABS(hflat)*rho0i*cpi
! 2. Sensible Heat Flux: Conduction when SST > T_air MUST remove energy from
ocean (< 0)
IF (TseaC .gt. TairC) THEN
  hfsen=-ABS(hfsen)*rho0i*cpi
ELSE
  hfsen=-hfsen*rho0i*cpi
ENDIF

```

This is a narrower, more mechanical fix than a full unit-mismatch or dynamic-SST-feedback correction – it guards against one specific failure mode (a sign flip producing spurious ocean warming from evaporation/conduction) rather than rewriting the bulk flux calculation. Whether this alone was sufficient to prevent the thermal runaway seen in early pre-alpha runs, versus other contributing factors (e.g. the vertical-coordinate/barotropic-transport fixes in `prepare_croco_forcing.py` described in Section 2 of `02_modeling_suite.md`), has not been isolated by a controlled test; both changes shipped together in the runs validated so far.

3. Nocturnal Boundary Cooling Mechanics

The resolution of `bulk_flux.F` restores physical nocturnal cooling. During daytime hours, solar flux (radsw) dominates, producing a positive net flux ($\text{shflux} \approx +400 \text{ W/m}^2$ to $+700 \text{ W/m}^2$) that warms the top 1 m – 3 m diurnal skin layer.

During night hours ($SW_{\downarrow} = 0$), radsw drops to zero. Net surface heat flux becomes strictly negative:

$$\text{shflux}_{\text{night}} = \text{shflux}_{\text{rlw}} + \text{shflux}_{\text{lat}} + \text{shflux}_{\text{sen}} \approx -180 \text{ W/m}^2 \text{ to } -320 \text{ W/m}^2$$

Air-Sea Thermodynamic Surface Heat Flux Balance

Shortwave ($SW_{\downarrow}$)

Solar Downward Radiation
(Heat Gain)

Net Longwave (LW)

Thermal IR Exchange
(Night Cooling)

Sensible Heat (H_{sen})

Turbulent Air-Sea Heat
Transfer

Latent Heat (H_{lat})

Wind Evaporative Loss
(Primary Cooling)

CROCO Upper Mixed Layer Integration (`bulk_flux.F` Fixed)

$$\text{shflux} = \text{radsw} + \text{shflux}_{\text{rlw}} + \text{shflux}_{\text{lat}} + \text{shflux}_{\text{sen}}$$

This negative nocturnal flux generates surface water density inversion ($\frac{\partial \rho}{\partial z} < 0$), triggering convective vertical mixing that cools the surface layer back toward equilibrium baseline temperatures. Confirming this quantitatively against satellite radiometry or buoy observations for the Western Mediterranean has not yet been done – see `04_empirical_validation.md` for what has actually been measured so far.

04. Empirical Validation: Current Status Across the 5 Western Mediterranean Regions

This document tracks which region/horizon/timestep combinations have actually been run to completion and passed the structural validation gate (`scripts/validate_marine_output.py`), as opposed to a single polished “benchmark” narrative. It intentionally does not include specific SST/salinity/current numeric results, buoy RMSE figures, or named-location anomaly case studies, because none of those have actually been pulled from a completed run’s validation JSON and cross-checked yet – see “What is not yet validated” below.

1. Regional Domain Specifications

Region ID	Extent (lon / lat)	Grid ($\xi_\rho \times \eta_\rho$)	Grid Points	Vertical Levels (N)
Balearic 1km	0.5–5.5° E, 37.5–41.5° N	501 × 401	200,901	32
Alboran 1km	–6.0– –1.0° E, 35.0 –37.5° N	501 × 251	124,203	32
Gulf of Lion 1km	2.0–6.5° E, 41.5–43.3° N	500 × 200	not recomputed for new bbox*	32
Tyrrhenian 1km	7.5–14.0° E, 38.0–44.5° N	650 × 651	391,379	32
Algerian Basin 1km	–1.0–8.5° E, 35.0–38.0° N	951 × 301	282,273	32

*Gulf of Lion’s extent and grid were changed on 2026-07-29 (bbox `latitude_max` reduced from 44.5° to 43.3°; see `02_modeling_suite.md` §4.4). The grid dimensions above (`xi_rho=500`, `eta_rho=200`) are confirmed directly from the regenerated CROCO grid’s NetCDF; the “Grid Points” figure used elsewhere in this suite has not been recomputed against the new bbox and is omitted here rather than restating the stale pre-resize number.

2. What Has Actually Passed the Validation Gate

The proven CROCO timestep configuration this development cycle is $\Delta t = 30$ s baroclinic / NDTFAST=45 barotropic substeps (matching the historically stable $\Delta t_{fast} \approx 0.667$ s from an earlier confirmed $\Delta t = 20$ s/NDTFAST=30 configuration). Runs are accepted only if `validate_marine_output.py` confirms `finite_fraction: 1.0` (no NaN/Inf/uninitialized cells) and all output variables fall within their configured physical ranges, before a `CROCO_SUCCESS` marker is written.

Region	6h gate	24h gate	72h gate
Balearic 1km	✓ passed	✓ passed	attempted, hit a WRF-forcing data-availability limit (see below), not yet re-run
Alboran 1km	✓ passed	✓ passed	attempted, failed on the same WRF-forcing limit
Gulf of Lion 1km	✓ passed	✓ passed	attempted, failed on the same WRF-forcing limit
Tyrrhenian 1km	✓ passed	not yet run (user deferred after 6h/72h decision)	attempted, hit the same WRF-forcing limit
Algerian Basin 1km	✓ passed	✓ passed	attempted, hit the same WRF-forcing limit

The 72h attempts across all 5 regions failed not from a CROCO numerical problem, but because the WRF atmospheric forcing dataset reused for that test only had 24 hours of `wrfout_d02_*` output available (confirmed directly via `gsutil ls`) against a `run_marine_simulation.py` check requiring `forecast_hours + 1` hourly files. The fix in progress is to run WRF fresh for each forecast horizon (via the corrected `daily_orchestrator.py --use-gcp-batch` path) rather than reusing an older, shorter WRF dataset.

Gulf of Lion's grid was regenerated on 2026-07-29 (bbox resize described in `02_modeling_suite.md` §4.4, new dimensions 500×200 , replacing the previous 451×301 grid). The 6h/24h passes recorded in the table above were run against the *old* grid; the new grid has not yet been re-run at any CROCO horizon. A 3-region validation test covering the new Gulf of Lion grid (alongside Alboran and Algerian) is in progress as of this writing – see Section 3 below.

3. What Is Not Yet Validated

To avoid repeating the previous version of this document's mistake, the following are explicitly **not yet done**, not just "not shown here":

- No SST/salinity/current/sea-level numeric results (min/max/mean) from any completed run have been pulled from GCS and recorded in this document.
- No comparison against SOCIB buoys, CMEMS satellite SST, or any other independent observation source has been performed.
- No wave height (H_s) validation has been done – SWAN has not yet been run to completion across all 5 regions in a single pipeline run. A live 6-hour-forecast attempt this cycle surfaced two real, since-fixed bugs (a silently-deep land-boundary substitution and a missing numerical scheme accommodation for geometrically constrained domains – see `02_modeling_suite.md` §§4.4–4.5) plus an unrelated orchestrator timeout-sharing bug that killed the run prematurely (see `05_cloud_deployment_and_ops.md` §1a). A subsequent 3-region

test (alboran_1km, gulf_of_lion_1km, algerian_1km) with the fixes applied got a mixed result: gulf_of_lion_1km failed fast on a genuine bbox problem (now fixed by a bbox resize), while alboran_1km/algerian_1km ran 3.5+ hours without completing before being orphaned when the run aborted – inconclusive, not a pass. A fresh 3-region test with the fully-fixed image is in progress as of this writing (Cloud Build ID a8b61b91-2302-4d24-81ad-70f64951922e, started 2026-07-29T14:41:36Z); its result is not yet known.

- No named-location physical anomaly (e.g. a specific channel wind-acceleration or upwelling case study) has been identified from real model output. Any such example in an earlier version of this document was illustrative, not measured.

4. C-Grid Dimension-Aware Masking

In 3D hydrodynamic solvers using staggered Arakawa C-grids, variables reside on distinct spatial sub-grids:

$$\text{Tracers } (T, S, \zeta) \in \text{Grid}_\rho(\eta_\rho, \xi_\rho), \quad U \in \text{Grid}_u(\eta_u, \xi_u), \quad V \in \text{Grid}_v(\eta_v, \xi_v)$$

If a validation script blindly applies $\text{Mask}_\rho(\eta_\rho, \xi_\rho)$ to velocity components on a different sub-grid, xarray can perform an unintended outer join across non-matching spatial dimensions, ballooning memory use. `validate_marine_output.py`'s `_apply_matching_mask` guards against this by confirming $\text{dims}(\text{Mask}) \subseteq \text{dims}(\text{DataArray})$ before masking. (The specific “reduces to 1.8 seconds” performance figure from an earlier version of this document was not independently re-measured for this rewrite and has been removed rather than repeated unverified.)

05. Cloud Deployment & Serverless Operations

This document describes the cloud-native, serverless execution framework developed for **PredSea** using **Google Cloud Platform (GCP) Batch**, **Spot VM instances**, and automated cost-protection mechanisms.

1. Containerized HPC Engine (croco-batch)

Rather than maintaining dedicated, static HPC server hardware, the entire numerical modeling environment—including compiled Fortran 90 binaries for CROCO 2.1.3 and SWAN 41.45, OpenMPI execution runtimes, NetCDF C/Fortran libraries, and Python spatial post-processors—is encapsulated in an immutable Docker container image (`simulation/marine/croco/Dockerfile.batch`).

```

+-----+
+
+               PredSea Unified HPC Container Architecture
+
+-----+
| Base OS: Ubuntu 22.04 LTS + OpenMPI 4.1 + gfortran / gcc
+-----+
| Compiled Binaries (one CROCO binary per region, built in a Dockerfile loop):
|   - /usr/local/bin/croco_balearic_1km, croco_alboran_1km, croco_algerian_1km, |
|     croco_gulf_of_lion_1km, croco_tyrrhenian_1km (region-specific LM/MM/N)
|   - SWAN binary (swan.exe/swanrun) is region-agnostic; bathymetry is what's
|     per-region – all 5 regions' SWAN bathymetry grids are COPY'd into the
|     image at build time (see docs/bathymetry-integration-guide.md)
+-----+
| Python Environment: Python 3.11 + NetCDF4 + xarray + SciPy + NumPy
+-----+
| Automation Scripts:
|   - run_marine_simulation.py  (Master entrypoint; --model croco or --model=
|     swan – `--model=both` is intentionally hard-disabled, see Section 2)
|   - prepare_croco_forcing.py / prepare_croco_bulk_forcing.py (forcing prep)
|   - validate_marine_output.py (Dimension-aware C-grid physical validator)
+-----+
+
+

```

The image is pinned by digest (not by mutable tag) in every real Batch submission, since digests can drift as the image is rebuilt: `europa-west1-docker.pkg.dev/predsea-api/predsea-simulations/croco-batch@sha256:f8313ce5d43624d055321ba769f7a7da315c6aad5c7e16c30ab10ba063953ba8`

This is the digest confirmed, this development cycle, to be built from a Dockerfile that compiles both CROCO and SWAN from source and bakes in per-region bathymetry for all 5 Western Mediterranean regions. This particular digest also carries two 2026-07-29 SWAN fixes (an explicit error when a boundary's walk-inward substitute exceeds 30% of that edge's grid size, and the addition of `PROP_BSBT` to SWAN's numerical scheme for geometrically constrained domains – both described in `02_modeling_suite.md` §§4.4-4.5) and the regenerated Gulf of Lion CROCO grid/binary (LM=498, MM=198, also §4.4). `submit_gcp_batch_simulation.py` refuses to submit a real (non-dry-run) job unless `--image-uri` contains an explicit `@sha256:` digest. Full end-to-end confirmation that SWAN actually runs successfully against real bathymetry for all

5 regions is still open – see 04_empirical_validation.md for the current, in-progress validation status.

1a. Orchestration Entry Point and the WRF → CROCO → SWAN Sequence

The daily/gate pipeline is not submitted by hand region-by-region. A single Cloud Build job invokes `scripts/daily_orchestrator.py --use-gcp-batch`, which drives the full sequence:

1. **Boundary/credential fetch:** ECMWF and CMEMS forcing are downloaded first. Copernicus, AEMET, and SOCIB credentials are pulled from Secret Manager at build time (`availableSecrets/secretEnv` in the Cloud Build config) rather than from a local `.env` file, since `humanintheloop/.env` is git-ignored and never reaches Cloud Build's source upload. Builds run as the Compute Engine default service account, not the legacy Cloud Build SA – that account is what needs `roles/secretmanager.secretAccessor`.
2. **Shared WRF run:** a single Spot/on-demand VM runs WRF once for atmospheric forcing. Every region's CROCO/SWAN job reads from this same WRF output – WRF is never run per-region.
3. **CROCO phase:** once WRF completes, a CROCO Batch job is submitted for every configured region simultaneously ($5 \text{ jobs} \times 16 \text{ vCPU c2d-highcpu-16} = 80 \text{ vCPU}$), and the orchestrator polls each job's real Batch state plus a `CROCO_SUCCESS` GCS marker until all regions finish.
4. **SWAN phase:** only after every region's CROCO job succeeds does the orchestrator submit SWAN as a second, separate phase across all regions, polling the same way for a `SUCCESS` marker.

Two phases instead of one combined submission exist for two independent reasons:

- `run_marine_simulation.py --model=both` is a hard, permanent `parser.error()` (exit code 2) – CROCO and SWAN must always run as separate Batch jobs, never combined in one container invocation.
- The project's `CPUS_ALL_REGIONS` quota is a single **global** 64 vCPU cap shared by the WRF VM and every Batch job across every phase – it is not a per-phase or per-region allowance. 5 regions' CROCO jobs alone ($5 \times 16 \text{ vCPU} = 80 \text{ vCPU}$) already exceed it, so in practice some regions' jobs queue for capacity rather than all starting immediately (a multi-minute `CPUS_ALL_REGIONS` quota wait per region has been observed in production runs). Splitting CROCO and SWAN into two sequential phases avoids also stacking all 10 region×model jobs at once (160 vCPU), but does not remove the queuing itself.

Any region is mandatory within its phase – a submission or completion failure for one region aborts the whole run rather than silently publishing a partial result. This is not just a design description: a 2026-07-29 3-region validation test was aborted mid-run when `gulf_of_lion_1km`'s SWAN job failed, and the other two regions' still-running SWAN jobs (`alboran_1km`, `algerian_1km`, each past 3.5 hours at that point) were orphaned as a result rather than left to finish – see 04_empirical_validation.md for the current status of that test.

Timeout budgets: the WRF VM timeout and the Batch-phase timeout are now decoupled

Until 2026-07-29, `daily_orchestrator.py` computed a single `timeout_hours` value from the forecast horizon ($\max(4.0, (\text{forecast_hours} / 24) * 1.25)$, flooring at 4.0h for short horizons) and reused it for two unrelated things: the WRF VM's own timeout, and the combined CROCO+SWAN Batch-phase timeout. This caused a real production incident: a live 6-hour-forecast test run was killed by the shared 4-hour ceiling while 3 of the 5 regions' SWAN jobs were still legitimately in progress – not hung, just needing more than 4 hours combined for CROCO+SWAN across all regions.

The fix adds a separate function, `batch_pipeline_timeout_hours()` (floor of 8.0h, scaling with forecast horizon), used only for the Batch phase and for each Batch job's own `maxRunDuration`, fully decoupled from the WRF VM's own timeout. `cloudbuild.6h-gate.yaml`'s own Cloud Build step timeout was also bumped from 6 hours (21600s) to 14 hours (50400s) – the old ceiling would otherwise have killed the entire build before the new, larger internal timeout could ever matter, since WRF (up to ~4h) plus the Batch phase (up to ~8h) can now total up to 12h in the worst case.

2. Dynamic Resource-Aware Compute Sizing & Regional Matrix

To prevent Out-Of-Memory (OOM) failures while minimizing compute expenditure, the submission orchestrator `scripts/submit_gcp_batch_simulation.py` provisions compute resources according to regional grid point densities:

$$\text{Grid Points} = \left(\frac{(\text{Lat}_{\max} - \text{Lat}_{\min}) \times 111,000}{H_{\text{res}}} \right) \times \left(\frac{(\text{Lon}_{\max} - \text{Lon}_{\min}) \times 111,000 \times \cos(\text{Lat}_{\text{mid}})}{H_{\text{res}}} \right)$$

Regional Deployment Specifications (Western Mediterranean Suite)

`calculate_resources()` in `submit_gcp_batch_simulation.py` auto-derives machine sizing from each region's grid point count, but its "small tile" and "medium tile" branches currently return identical values – so every real submission this cycle has passed explicit CLI overrides to get the proven configuration below, rather than relying on the auto-sizing:

Region ID	Grid Points	Machine Type	MPI Ranks	Compute Spec	Provisioning
Balearic 1km	200,901	c2d-highcpu-16	16	16 vCPU / 32 GiB	STANDARD
Alboran 1km	124,203	c2d-highcpu-16	16	16 vCPU / 32 GiB	STANDARD
Gulf of Lion 1km	121,649*	c2d-highcpu-16	16	16 vCPU / 32 GiB	STANDARD
Tyrrhenian 1km	391,379	c2d-highcpu-16	16	16 vCPU / 32 GiB	STANDARD
Algerian 1km	282,273	c2d-highcpu-16	16	16 vCPU / 32 GiB	STANDARD

*Gulf of Lion's Grid Points figure above predates the 2026-07-29 bbox resize (02_modeling_suite.md §4.4) and has not been recomputed against the new bbox (latitude_max now 43.3° instead of 44.5°); it is left as-is here rather than replaced with an invented number, but should not be treated as current. Machine sizing (16 vCPU) was set via explicit CLI override in any case, not derived from this figure.

STANDARD (on-demand) provisioning was chosen deliberately over SPOT for these validation runs: SPOT capacity draws from the same single global CPUS_ALL_REGIONS quota pool (64 vCPUs for the whole project, covering the WRF VM and every Batch job together – see Section 1a), and deadline-critical gate testing needs guaranteed capacity rather than preemption risk. SPOT remains the CLI default in `submit_gcp_batch_simulation.py` and is the intended cost-saving mode once the pipeline is stable enough to tolerate retries.

3. Spot VM Preemptibility & Cost Safety Traps

SPOT VM pricing is **70%-90%** cheaper than on-demand, and is the long-term target once this pipeline no longer needs deadline-guaranteed capacity (see Section 2). The fault-tolerance mechanisms below apply regardless of provisioning model:



- 1. Automated Self-Deletion Traps:** Compute nodes automatically self-destruct upon task exit, eliminating idle VM billing risks.
- 2. In-Cloud Validation Gate:** All assertions (`validate_marine_output.py`) run in-cloud directly inside the container before storage sync, requiring zero local downloads.
- 3. Run-Scoped Immutability:** Forecast outputs are committed to immutable GCS storage paths: `gs://predsea-daily-outputs-test/predictions/YYYY-MM-DD/runs/[RUN_ID]/`

06. Conclusion & Engineering Roadmap

This document summarizes the technical achievements of the **PredSea** oceanographic forecasting architecture and outlines the future engineering roadmap.

1. Summary of Current Architectural State

PredSea's working hypothesis is that high-resolution (1 km) regional ocean modeling does not require expensive, dedicated supercomputer infrastructure – that serverless GCP Batch orchestration, parallel process-level Python pre-processing, and the CROCO/SWAN/WRF numerical engines can deliver it instead. As of this development cycle:

1. **High-Resolution Coastal Granularity:** The 5 Western Mediterranean 1 km CROCO grids are defined and have compiled binaries; validated runs to date confirm numerical stability at 6h and 24h horizons (see `04_empirical_validation.md` for the exact per-region status), not yet a full operational forecast product.
 2. **Compute Orchestration:** CROCO and SWAN run as parallel per-region GCP Batch jobs sharing one upstream WRF run; `daily_orchestrator.py` now chains WRF → regional CROCO+SWAN submission → real Batch-status polling automatically (see `05_cloud_deployment_and_ops.md`). A 2026-07-29 production incident (a shared timeout budget between the WRF VM and the combined CROCO+SWAN Batch phase killed a live 6h test while 3 of 5 regions were still legitimately running) led to decoupling those two timeouts – see `05_cloud_deployment_and_ops.md` §1a. Specific pre-processing/runtime speed numbers have not been re-measured since this pipeline changed and are not repeated here to avoid restating stale figures.
 3. **Cost:** STANDARD (on-demand) provisioning is currently used deliberately for reliability during validation, not SPOT; a verified, current per-day cost figure for the full 5-region suite has not yet been compiled. SPOT is the intended cost-saving target once the pipeline is stable (see `05_cloud_deployment_and_ops.md`, Section 2).
 4. **Physical & Empirical Accuracy:** A verified fix exists for one specific bulk-flux sign-convention bug (`03_thermodynamics_and_fluxes.md`, Section 2). No comparison against SOCIB buoys, satellite SST, or other independent observations has been performed yet – this remains open work, not a completed accomplishment.
 5. **Wave Engine Migration (in progress):** SWAN is being replaced project-wide by Wave-Watch III (WW3) after exhaustive tuning ruled out every SWAN parameter as the cause of persistent non-convergence in 3 of the 5 regions (see `02_modeling_suite.md`, Section 5). A 5-region 6h WW3 breadth test has passed with a workload-proportional core allocation; a WRF-forced 24h, 5-region test is the next validation gate, currently blocked on a WRF domain-coverage fix. See `docs/agent-handoff-ww3-wrf-migration-2026-08-05.md` for exact current status.
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2. Technical & Strategic Roadmap

Milestone A: Multi-Day Diurnal Thermal Skin Layer Tracking

The CROCO 24-hour stability gate has passed for most regions (see 04_empirical_validation.md), but multi-day forecasting (96 – 120 hours) during summer marine heatwaves requires higher vertical resolution in the upper 1 meter ocean skin layer:

- **s-Coordinate Refinement:** Increase vertical levels from $N = 32$ to $N = 40$ or $N = 50$, increasing surface stretching ($\theta_s = 8.0$) to place 5 vertical layers within the top 1 meter.
- **Diurnal Warm-Layer Modeling:** Integrate explicit Cool-Skin / Warm-Layer parameterizations (e.g. Fairall et al. COARE scheme) to track diurnal surface warming peaks ($+1.5^\circ\text{C} - 2.5^\circ\text{C}$ afternoon spikes) and nocturnal mixing decay.

Milestone B: Full Operational CROCO-Wave Wave-Current Coupling

Expand two-way standalone runs into active three-way dynamic OASIS3-MCT coupled cycles. Written against SWAN below; given the SWAN→WW3 migration described in 02_modeling_suite.md Section 5, this milestone's counterpart is expected to become WW3 once the migration completes, not SWAN – CROCO's own OASIS coupling toolbox is solver-agnostic on the wave side, so this does not change the milestone's feasibility, only which binary it targets:

- **Wave Radiation Stress Feedback:** Dynamically pass SWAN S_{xx}, S_{xy}, S_{yy} radiation stress gradients into CROCO to drive wave-induced longshore currents, wave setup in harbors, and wave-current bottom friction enhancement.
- **Current Refraction Feedback:** Pass CROCO 1 km hourly surface currents back into SWAN to compute Doppler-shifted wave refraction in high-current channels (e.g., Ibiza-Formentera Freus channel).

Milestone C: Automated Captain-Facing Decision APIs & Alerting

Bridge raw numerical model outputs directly to operational maritime decision tools:

- **Vessel Response Threshold Engine:** Translate 1 km wave spectra, wind against current vectors, and cross-channel steepness into vessel-class safety statuses (favorable, workable, conservative, restricted) for small ($< 12\text{ m}$), medium ($12 - 24\text{ m}$), and large ($> 24\text{ m}$) motor and sailing yachts.
- **Automated Artifact Dispatch:** Automatically compile daily briefing maps, WhatsApp captain advisories, and LinkedIn operational summaries upon forecast completion.
- **FastAPI / Deck.gl Production Endpoints:** Serve real-time oceanographic vector fields and wave condition layers to mobile and web dashboards at sub-second latencies.

3. Concluding Remarks

This document set describes an architecture in active development, not an operational product. The core design choices – one shared WRF run feeding independent per-region CROCO/SWAN Batch jobs, run-scoped GCS outputs, structural validation gates before publication – are in place and have started producing real, passing 6h/24h regional runs. What remains before this can honestly be called a validated forecasting platform for maritime operators is exactly what Section 2's roadmap and the open items in 05_cloud_deployment_and_ops.md describe: multi-day horizon coverage across all 5 regions, real coupling between CROCO and SWAN,

independent observational validation, and the reliability/observability hardening the architecture review identified. Overstating current status has been a recurring problem in earlier drafts of this document set; this rewrite is intended to fix that, not to replace one round of overclaiming with another.